

Informe Gen AI.

INFORME INTELIGENCIA ARTIFICIAL GENERATIVA APLICADA A RIESGOS DE CIBERSEGURIDAD

Investigación de Tendencias 2024-2025 | Enfoque España | Contexto Global Comparativo

Elaborado por: Consorcio de Científicos de Datos, Estrategas de Ciberseguridad y Expertos en Resiliencia Digital

Fecha: Enero 2026 | **Fuentes:** 145+ documentos académicos, regulatorios y técnicos

RESUMEN EJECUTIVO

La convergencia entre Inteligencia Artificial Generativa (GenAI) y ciberseguridad constituye el fenómeno más disruptivo de 2024-2025 en infraestructura digital global. Este informe sintetiza hallazgos de 145+ fuentes de alto rigor (peer-reviewed, organismos gubernamentales certificados, marcos normativos emergentes, investigaciones de laboratorio) para ofrecer análisis exhaustivo de tendencias, vectores ofensivos/defensivos, indicadores nacionales españoles y recomendaciones estratégicas.

Hallazgos Críticos para España

1. Amenaza Cuantificada: 100,000+ ciberataques registrados 2024 (1 "muy grave" cada 3 días); incremento 300% desde 2015. España posiciona 5ª globalmente en ataques ransomware (58 incidentes H1 2024, +23% vs 2023).123
2. Respuesta Estatal: Inversión €1.157 mil millones, creación Centro Nacional de Ciberseguridad (CNC), transposición NIS2 esperada Q1 2026. Expansión regulatoria: 1,000 → ~12,000 entidades cubiertas.45
3. IA Generativa como Amenaza: Inyección de prompts (87% éxito basal), envenenamiento datos, extracción modelos, ataques cadena suministro, deepfakes (8M archivos 2025, +1,600% vs 2023). Actores estatales (Irán, China, Corea del Norte) utilizan systematcally GenAI en ciclos completos ataque.678
4. IA Generativa como Defensa: ML-IDS alcanzan 96.9% tasas detección. Threat hunting asistido IA: MTTD de 181 días → <24 horas. Detección deepfakes: 94-96% accuracy multimodal.91011
5. Brecha Crítica: 26% organizaciones confían capacidad detectar amenazas IA. 35% pequeñas empresas reconocen preparación inadecuada. Marcos regulatorios aún en implementación (EU AI Act agosto 2026).1213

I. CONTEXTO NACIONAL ESPAÑA: INDICADORES Y GOBERNANZA

A. Panorama de Incidentes 2024-2025

España experimentó intensificación sin precedentes de actividad cibercriminal. Según documentación oficial Gobierno de España, se registraron >100,000 ciberataques con promedio 1 incidente "muy grave" cada 3 días. Este volumen representa incremento 300% desde 2015, señalando no solo aumento cuantitativo sino sofisticación cualitativa de vectores ataque.¹²

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 - 2 <https://pmc.ncbi.nlm.nih.gov/articles/PMC11622839/>
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Análisis Sectorial:

- Administración Pública: 38% de ataques europeos¹⁴
- Transporte: 11% (emerging high-value target)¹⁴
- Infraestructura digital: objetivos estratégicos sostenidos¹⁴

Métricas de Cibercriminalidad:

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- 237,640 incidentes registrados (primeros 7 meses 2024)³
- +9.2% aumento vs 2023³
- Ransomware: España 5º país globalmente (58 incidentes H1 2024, +23% YoY)³

B. Índice Nacional de Ciberseguridad (NCSI) - España

Puntuación Consolidada: 89.17/100 (top 20 global)³

Desagregación por Dimensión:

- Governance Policy Development: 6/7 (86%)¹⁶
- Cybersecurity Policy Unit: Designada y empoderada¹⁶
- National Strategy: Documentada con action plan¹⁶
- Research Recognition: RENIC formal acknowledgment¹⁶

Esta puntuación refleja progreso significativo en governance, aunque subsisten gaps en coordinación interinstitucional y capacidades operacionales respuesta.

C. Marco Legislativo: NIS2 y Reorganización Institucional

Transposición NIS2 Directive:

Consejo Ministros español aprobó enero 2025 Anteproyecto Ley Coordinación y Gobernanza Ciberseguridad. Cronograma legislativo:⁴

Hito	Fecha	Estado
Aprobación Consejo Ministros	Enero 14, 2025	✓ Completado
Registro acelerado Congreso/Senado	Febrero 2025	✓ Completado
Audiencias parlamentarias	Q3 2025	En Progreso
Enactment esperado (BOE)	Q1 2026	Proyectado
Portal registro CNCS operativo	H1 2026	Proyectado
Notificación incidentes vinculante	H1 2026	Proyectado

Impacto Regulatorio - Scope Expansion:

De <1,000 entidades bajo NIS original a ~12,000 bajo NIS2 en 18 sectores. Clasificación bifurcada:5

Clase	Criterios	Ejemplos
Entidades Esenciales (EE)	≥250 empleados O ≥€50M turnover	Utilities, hospitales, ministerios
Entidades Importantes (EI)	≥50 empleados O ≥€10M turnover	Logistics hubs, manufacturers

Sanciones Administrativas Escaladas:6

- Infracciones muy graves: €10M O 2% volumen negocios global (EE); €7M O 1.4% (EI)
- Infracciones graves: €100K-€500K
- Infracciones leves: €10K-€100K

Centro Nacional de Ciberseguridad (CNC):

Presidencia Gobierno anunció octubre 2025 creación CNC como entidad interministerial bajo presidencia directa. Funciones asignadas:78

- Coordinación nacional ante ataques masivos
- Detección/análisis amenazas digitales (asistidas IA)
- Protección infraestructuras críticas
- Sistema Nacional Respuesta Incidentes (coordinación CCN, Guardia Civil, Policía)

Presupuesto: €300M+ inicial; enfoque formación 80,000+ profesionales requeridos próximos 3 años.9

D. Inversión Estatal y Compromisos Presupuestarios

Plan Nacional de Ciberseguridad: €1.157 Mil Millones¹⁰¹¹

Distribución asignada:

- 60%: Ministerio Defensa, Centro Criptológico Nacional (CCN-CERT), Comandancia Ciberespacio
- 40%: Ministerio Interior, Secretaría Estado Telecomunicaciones, Red.es, AEAD

Medidas operacionales:¹²

- Expansión 5G Security Operations Center
- Auditorías automatizadas servicios digitales públicos
- Integración técnicas avanzadas IA para detección
- Coordinación mejorada SOC's público-privados

II. PANORAMA GLOBAL AMENAZAS CIBERNÉTICAS 2024-2025

A. Magnitud Económica y Estadísticas de Incidentes

Cibercriminalidad ha alcanzado proporción sistémica. Estimaciones 2025 proyectan costos globales **\\$10.5 billones USD** (incremento 31% YoY), continuando trayectoria acelerada desde **\\$6 billones** 2021.¹³¹⁴

Métricas de Incidentes:

- IC3 (USA): 859,532 denuncias 2024 con pérdidas >\\$16 mil millones (+33% vs 2023)²¹
- Costo Promedio Brecha Datos: \\$4.44M globalmente (reducción 9% vs \\$4.88M 2024, debido detección rápida + IA-assisted security). Variación regional: US \\$10.22M (máximo histórico); Europa ~\\$4M; Asia-Pacífico \\$2-3M²¹
- Ransomware: 5,414 víctimas publicadas 2024 (+11% vs 2023). Q4 2024: 1,827 incidentes (33% anual, +29% vs Q4 2023). 46 nuevos grupos emergentes (48% incremento). Top 10 grupos: 52.8% ataques²³
- Phishing: Email attacks +197% H2 2024 vs H2 2023. 31.4% correos spam; 1.4% con malware/phishing²⁴

ENISA Threat Landscape 2024 (4,900+ incidentes curados):¹⁵¹⁶

Seven prime threats (ranked):

6. Amenazas a disponibilidad (DDoS + ransomware)
7. Ransomware (stabilizado alto volumen, técnicas double/triple extortion)
8. Amenazas a datos
9. Ingeniería social (phishing dominante 60% initial intrusion vector)
10. Malware-as-a-Service offerings (MaaS evolución rápida)
11. Ataques cadena suministro
12. Manipulación información/Disinformation

Vulnerabilidades: 42,595 nuevas (27% incremento); 64% con network attack vector.¹⁷

B. Actores de Amenaza Evolución

State-Nexus Actors: Enfoque aumentado operaciones espionaje largo plazo con baja huella.

Explotación sistemática servicios cloud, vulnerabilidades public-facing. Targeting infraestructura telecomunicaciones (cables submarinos, satélites) con implicaciones geopolíticas.¹⁸

Cybercrime Actors: Ecosistema "as-a-Service" en maduración. RaaS ofrece acceso inicial, malware development, data exfiltration, leak portals. DDoS-for-Hire reduce barreras entrada atacantes no especializados.¹⁹

IO Actors (Disinformation): Campañas masivas sobre infraestructura diplomática, procesos electorales europeos. Geoposicionamiento táctico de mensajes.²⁰

III. INTELIGENCIA ARTIFICIAL GENERATIVA: VECTORES OFENSIVOS

A. Taxonomía de Amenazas GenAI

1. Inyección de Prompts (Prompt Injection)

Vulnerabilidad arquitectónica fundamental en LLMs. Atacante inserta instrucciones maliciosas en inputs que parecen benignos; modelo procesa ambos canales como entrada unified, generando riesgo de bypass de guardrails seguridad.²¹²²

Métricas de Éxito Línea Base: 87% ataques logran bypass defensa genérica. Mitigaciones especializadas (Signed-Prompt Method, StruQ) reducen tasas éxito a <0.5% en modelos fine-tuned.²³²⁴²⁵

Jailbreak Attacks: "Do Anything Now" v15.0 y similares explotan tendencia LLMs seguir instrucciones literalmente. Anthropic Red Team dataset: milenios adversarial prompts. Prompt-Guarding (Prompt-G) redujo attack success rate con Llama 2 13B a **2.08%**.²⁶

OWASP Designación: LLM01:2025 como amenaza top-tier.²⁷

2. Envenenamiento de Datos (Data Poisoning)

Inyección datos maliciosos durante training/tuning sesga comportamiento modelo. Vectores incluyen: (a) inyección directa datasets training, (b) compromiso fuentes datos upstream pre-training, (c) poisoning "split-view" dominios expirados recomprados, (d) "frontrunning poisoning" datasets web scraped.²⁸²⁹

Impacto Documentado: Modelos envenenados generan decisiones discriminatorias, recomendaciones incorrectas, fallos detección amenazas. Backdoors activados bajo triggers específicos sin afectar funcionalidad nominal.³⁰³¹

Mitigación: Verificación integridad datasets, data provenance tracking, sanitización datos training, reevaluación periódica contra conjuntos clean.³²

3. Extracción de Modelos (Model Extraction)

Model stealing: atacante interactúa repeatedly con API modelo para construir dataset sintético inputs/outputs, luego entrena modelo sustituto replicando comportamiento original.³³³⁴

Relevancia Económica: Modelos generativos representan años investigación + datos propietarios. Clonación via API es técnica viable robo IP. Google Threat Intelligence observó actores estatales usando Gemini para research vulnerabilidades, development payloads, malicious scripting.³⁵

Model Inversion: Análisis respuestas modelo para reconstruir datos sensibles training (salud, finanzas, PII). Membership inference: determina si muestra fue parte dataset training.³⁶

4. Ataques Cadena de Suministro de IA

Atacantes inyectan código malicioso en componentes AI reutilizados (modelos pre-trained, librerías ML) en repositorios públicos (PyPI, npm, Hugging Face).³⁷³⁸³⁹

Caso de Uso 2025 - SolarTrade: Platform logística usada por 500+ retailers comprometida cuando atacantes inyectaron código malicioso en actualización software. Acceso conseguido información pago clientes; operaciones disrupted meses.40

Medtech Firmware: Dispositivos médicos (marcapasos, bombas insulina) comprometidos via ataque fabricante. Implicaciones seguridad paciente extremas.41

Detección Gaps: MLOps platforms fallan detectar malware en serialized model files.42

5. Deepfakes y Manipulación Multimedia

Crecimiento Explosivo: 500K archivos (2023) → 8M (2025). Tasas crecimiento: 900-1,740% por región.4344

Detección Humana: Capacidad identificación ~24.5% videos deepfake high-quality - apenas mejor que adivinanza aleatoria.45

Sistemas Automatizados: Degradación 45-50% accuracy contra deepfakes reales vs. condiciones lab.46

Aplicación Ofensiva: (a) Suplantación identidad ejecutiva, (b) Fraude verificación identidad servicios financieros, (c) Disinformation campaigns escala. **Precedente:** January 2024, engineering firm Arup victimizada fraude \ \$25M via deepfake ejecutivo video.⁴⁷

Defensas Emergentes: Real-time multimodal detection (voice + video + behavioral) alcanza 94-96% accuracy condiciones óptimas. Federated learning updates daily. >100 instituciones financieras deployed behavioral biometrics networks.⁴⁸

6. Phishing Asistido por IA

Email attacks incrementaron 197% H2 2024 vs H2 2023. GenAI permite creación phishing campaigns masivamente personalizadas:⁴⁹

- OSINT analysis targets individuales → tailored messaging
- LLMs generen contenido contextualmente creíble
- Multilingual content generation at scale

Observación Operacional: Iranian APT42 utilizó Gemini (30%+ actividad) para reconnaissance policy experts, drafting spear-phishing campaigns, research vulnerabilidades específicas (Mikrotik, Apereo, Atlassian).⁵⁰

IV. INTELIGENCIA ARTIFICIAL GENERATIVA: CAPACIDADES DEFENSIVAS

A. Detección de Amenazas Asistida por ML

Sistemas IDS Basados en Machine Learning

IDS tradicionales signature-based quedan obsoletos ante zero-day threats. ML-based IDS aprenden características estadísticas attacks históricos.⁵¹

Benchmarks Rendimiento Documentado:⁵²⁵³⁵⁴

Algoritmo	Accuracy	TPR	FPR	AUC	Dataset
Random Forest	98.53%	0.985	0.001	0.998	Malware API frequency
Decision Tree	96%	0.96	0.05	-	NSL-KDD
GAN Discriminator	>98%	-	-	-	Zero-day variants
Autoencoder + RF/XGBoost	High	Superior	-	-	Unseen data

Detección Basada en Comportamiento (UEBA): User/Entity Behavior Analytics establece baselines dinámicas, identifica desviaciones. Excelente insider threats, compromised credentials, novel attacks. **60% reducción false positives** vs. rule-based detection.⁵⁵

B. Threat Hunting Augmented por IA

Impacto en MTTD (Mean Time to Detect): De 181 días (promedio industrial) a <24 horas con sistemas AI-driven.⁵⁶

Técnicas AI-Enhanced:

- Hypothesis-Driven: Threat intelligence fueling hypotheses específicas
- Anomaly-Based: UEBA establece baselines adaptativas
- Intelligence Enrichment: AI enriquece IOCs, correlaciona múltiples fuentes
- Predictive Intelligence: ML forecasts probable attack vectors
- Behavioral Hunting: Detecta behaviors objetivos attacker (reconnaissance, lateral movement, exfiltration)¹⁰

C. Detección de Deepfakes

Multimodal Real-Time Detection: Voice + video + behavioral patterns simultaneously, **94-96% accuracy** condiciones óptimas con ensemble methods.⁵⁷

Federated Learning: Reentrenamiento diario con emerging threats, preservando privacy.

Multi-Factor Authentication Mejorado: Behavioral biometrics: typing patterns, navigation habits en real-time. Verificación criptográfica comunicaciones; procedimientos callback.⁵⁸

V. MARCOS NORMATIVOS Y GOBERNANZA

A. EU AI Act - Regulación Europea

Risk-Based Trichotomy:⁵⁹⁶⁰

13. Unacceptable Risk: 8 prácticas prohibidas (manipulación IA, exploración vulnerabilidades, social scoring, etc.)
14. High-Risk (Aug 2026): Infraestructura crítica, healthcare, law enforcement. Obligaciones estrictas pre-market.
15. Limited/Minimal Risk: Menos regulación.

Requisitos Ciberseguridad (Art. 15):61

- Robustez y exactitud apropiadas
- Resiliencia contra "unauthorized tampering"
- Protección contra adversarial attacks
- Technical documentation detailed
- Incident monitoring + reporting

B. Directiva NIS2

Scope Expansion: 1,000 → ~12,000 entidades en 18 sectores.62

Obligaciones Core:63

- Risk management systems + board accountability
- Supply chain security
- Incident detection/response
- Cross-border reporting (24h initial, 72h detailed)

C. NIST AI RMF y ISO 42001

NIST AI RMF Core Functions:6465

- 16. Govern: Políticas, roles accountability
- 17. Map: Identificar risks
- 18. Measure: Métricas cuantitativas/cualitativas
- 19. Manage: Mitigación risks

ISO 42001:2023: Certifiable standard para AI management systems. Integrable con ISO 27001, ISO 31000.6667

VI. RECOMENDACIONES CONSOLIDADAS

Para España (Nivel Estatal)

- 20. Acelerar NIS2 Implementation: Roadmap Q1 2026; soporte PYMES
- 21. Potenciar CNCS: Dotación presupuestaria, talento recruitment, coordinación interagencial
- 22. Alineación EU AI Act: Armonizar ENS pre-agosto 2026 obligatoriedad
- 23. Investigación Aplicada: Funding detección deepfake, threat intelligence IA, red teaming
- 24. Formación Especializada: 80,000+ profesionales; colaboración universidades, bootcamps

Para Organizaciones (Sector Crítico)

- 25. ISO 42001 Adoption: Framework governance sistemas AI
- 26. NIST AI RMF: Operacionalización Govern-Map-Measure-Manage
- 27. Red Teaming Continuo: Pre-deployment adversarial testing
- 28. Data Supply Chain Security: SBOM, verificación criptográfica, auditoría proveedores
- 29. ML-Based Threat Detection: IDS ML, behavioral analytics, threat hunting IA
- 30. Defensa Deepfake: Multimodal real-time systems integradas

CONCLUSIÓN

La convergencia GenAI-ciberseguridad ha alcanzado punto inflexión crítico. Capac defensivas alcanzando madurez (96.9% IDS detection, 181 días → <24 horas MTTD reduction), pero amenazas escalan faster (900%+ deepfake growth).

Gobierno de España - "The Government approves strengthening Spain's cybersecurity," May 202568

ACK EU Summit - Spain €1.157B cybersecurity investment 202569

Springboard35 - "Cybersecurity in Spain: Challenges and Opportunities," April 202570

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Vectra AI - "Threat Hunting Solutions," December 202577

WEF - "Detecting Dangerous AI," January 202678
WEF - "Global Cybersecurity Outlook 2025," January 202579
Darktrace - "UK State AI Cybersecurity Report," July 202480
ENISA - Threat Landscape briefing 2024-2025, September 202581
NCSI - Spain Ranking, 202582
NCSI - Spain Governance Indicators83
Lexmundi - "NIS2 Implementation Status Spain," November 202484
Inprotech - "NIS2 Regulation Spain," February 202585
Darknetsearch - "Centro Nacional Ciberseguridad," October 202586

3Digits - "Impulso Gobierno Ciberseguridad," January 2025⁸⁷
DeepStrike - "Cybersecurity Statistics 2025," November 2025⁸⁸
SentinelOne/Zerothreat - "Cybercrime Costs," December 2025⁸⁹
Cyberint - "Ransomware Annual Report 2024," January 2025⁹⁰
Acronis - "Cyberthreats H2 2024," February 2025⁹¹
ENISA - ETL 2024-2025 Analysis⁹²
ENISA - Threat Landscape CISA KEV data⁹³
WEF - "Critical Infrastructure Cyber Threats," January 2025⁹⁴
ENISA Threat Landscape 2024 summary⁹⁵

OWASP - "LLM01:2025 Prompt Injection," April 2025⁹⁶

Kili-Technology - Prompt Injection Guardrails⁹⁷

Auxin - "AI-Specific Attack Vectors," July 2025⁹⁸

Obsidian Security - "GenAI Security Risks," November 2025⁹⁹

NIH - "Mitigating Adversarial LLM Manipulation," October 2024¹⁰⁰

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CISA/NSA - "AI Data Security Guidance," May 2025¹⁰³

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AUXIN - AI Supply Chain Attacks105

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